MSC 201 **51270**

Unit 2 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. A proposed event probability is 1.15. What is wrong?	A. Probabilities must always equal 0.5. B. Only complements may exceed 1. C. All probabilities must be integers. D. A probability cannot exceed 1. 10 pts
2. Ten independent trials each have success probability 0.3. X counts successes. What makes a binomial model appropriate?	A. The expected count is an integer. B. Fixed trial count, two outcomes, independence, and constant success chance. C. Success probability exceeds 0.5. D. Every trial must succeed. 10 pts
3. Independent tasks each succeed with probability 0.9. Which shortcut gives at least one failure among three tasks?	A. 1 minus the probability exactly one succeeds. B. 1 minus the probability all three succeed. C. The probability all three fail. D. 0.9 plus 0.9 plus 0.9. 10 pts
4. An independent fair coin has just produced five tails. What is the chance of heads next?	A. 1. B. 1/2. C. Less than 1/2 because tails is favored. D. Greater than 1/2 because heads is due. 10 pts
5. A teacher draws cards without replacement until the first marked card. Why is the draw number generally neither binomial nor geometric?	A. Stopping at the first success always makes the count binomial. B. The trial count is not fixed, and the chance can change between draws. C. The draw number is continuous. D. A success has to be a desirable outcome. 10 pts
6. A model assigns a 5% chance to 4 failures. One session has 4 failures. What follows?	A. Its probability must now be 100%. B. All future sessions must have 4 failures. C. The observation is uncommon but consistent with the model. D. The model is disproved by this one allowed outcome. 10 pts

Test continues on the other side of the sheet.

MSC 201 **51270**

Unit 2 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. Two distinct terminal paths describe exactly one head in two tosses. How is their combined probability found?	A. Multiply the two path probabilities. B. Ignore one because both describe one head. C. Subtract the smaller from the larger. D. Add the two path probabilities. 10 pts
8. A technician stops testing when a device first works. If the variable counts trials through the first success, which attempts are included?	A. The failed attempts and the attempt after the success. B. Only the failed attempts. C. Only the first successful attempt. D. Every failed attempt and the first successful one. 10 pts
9. A count variable has expected value 0.4. What does that mean?	A. The model is invalid. B. Its long-run average can be 0.4 despite integer observations. C. Exactly 40% of sessions must contain one event. D. A single session must contain 0.4 events. 10 pts
10. Which design permits an exact binomial count model with independent randomized draws?	A. Replacing and remixing each drawn token. B. Drawing without replacement until the box is empty. C. Always stopping at first red. D. Drawing most of a small population without replacement. 10 pts

51270

TEST PAGE 1

1.

D.

10 pts

2.

B.

10 pts

3.

B.

10 pts

4.

B.

10 pts

5.

B.

10 pts

6.

C.

10 pts

TEST PAGE 2

7.

D.

10 pts

8.

D.

10 pts

9.

B.

10 pts

10. A.

10 pts

MSC 201 **58181**

Unit 2 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. A small prize is common and a large prize is rare. Why is the midpoint between the prizes generally a poor estimate of the expected payout?	A. Only the larger prize determines the mean. B. It gives the two prizes equal weight despite their different chances. C. The expected payout must equal the larger prize. D. The small prize does not count toward the average. 10 pts
2. P(A)=0.30 and P(A given B)=0.30, with P(B)>0. What does this support?	A. A and B are mutually exclusive. B. B guarantees A. C. A is impossible. D. A and B are independent. 10 pts
3. Among 20 tablet users, 12 also use laptops. What is P(laptop given tablet) in this sample?	A. 8/20. B. 12/20. C. 12 divided by all college students. D. 20/12. 10 pts
4. A device is tested a fixed number of times, but each repair after a failed trial improves its chance of passing. Why is a simple binomial model questionable?	A. The result is a count. B. The pass chance depends on previous outcomes. C. The test records successes and failures. D. The number of trials is fixed. 10 pts
5. On a die, A is even and B is greater than 3. Which results satisfy at least one condition?	A. 4, 6 only. B. All six results. C. 2, 4, 5, 6. D. 1, 3 only. 10 pts
6. X counts broken cables in independent checks. Can broken be called success in a binomial model?	A. Yes; success means the counted outcome. B. No; success must be desirable. C. Only if more than half are broken. D. Only if no cables are broken. 10 pts
7. The success chance improves after each failed attempt because the process learns. Why is a geometric model questionable?	A. Failures are counted. B. There is a stopping rule. C. The success probability is not constant. D. The first success occurs eventually. 10 pts

Test continues on the other side of the sheet.

MSC 201 **58181**

Unit 2 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

8. Which expression computes P(X>0) for a nonnegative count X?	A. P(X=0). B. 1-P(X=1). C. The mean of X. D. 1-P(X=0). 10 pts
9. Given A, the chance of B is 0.7. What is the chance of not B given A?	A. Cannot be found unless A is independent of B. B. 0.3. C. 1.7. D. 0.7. 10 pts
10. A sample is drawn without replacement from a box containing marked and unmarked cards. When must the sample contain a marked card?	A. Only if every card in the box is marked. B. Whenever one marked card is present in the box. C. When the sample size exceeds the number of unmarked cards. D. Whenever the sample size is less than the number of unmarked cards. 10 pts

58181

TEST PAGE 1

1.

B.

10 pts

2.

D.

10 pts

3.

B.

10 pts

4.

B.

10 pts

5.

C.

10 pts

6. A.

10 pts

7.

C.

10 pts

TEST PAGE 2

8.

D.

10 pts

9.

B.

10 pts

10.

C.

10 pts

MSC 201 **27898**

Unit 2 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. A small prize is common and a large prize is rare. Why is the midpoint between the prizes generally a poor estimate of the expected payout?	A. The expected payout must equal the larger prize. B. The small prize does not count toward the average. C. It gives the two prizes equal weight despite their different chances. D. Only the larger prize determines the mean. 10 pts
2. An inspector checks a fixed batch of independent items with the same defect chance. Which variable fits a binomial model?	A. The average time needed to check an item. B. The exact mass of the first item. C. The attempt on which the first defect is found. D. The number of defective items found. 10 pts
3. Two independent retry procedures each have a constant chance of success. Which tends to have the shorter wait for the first success?	A. Whichever procedure happened to fail first. B. The one with the lower success chance per try. C. The one with the higher success chance per try. D. They always have the same typical wait. 10 pts
4. An event is described as drawing a heart or a king. Does the king of hearts qualify?	A. Yes; the statistical OR includes the overlap. B. No; OR always excludes overlap. C. Only if the card is drawn twice. D. Only if hearts and kings are independent. 10 pts
5. Six tokens are drawn without replacement from a small mixed box. What challenges an exact binomial model?	A. The number of draws is fixed. B. The count is an integer. C. Draws change the composition and are dependent. D. There are only two colors. 10 pts
6. Half of tablet users use laptops, and half of non-tablet users use laptops. What does this suggest in these data?	A. Laptop use does not depend on tablet-use status. B. The events are mutually exclusive. C. Every student uses both devices. D. Every student uses exactly one device. 10 pts
7. P(A)=0.30 and P(A given B)=0.30, with P(B)>0. What does this support?	A. B guarantees A. B. A is impossible. C. A and B are mutually exclusive. D. A and B are independent. 10 pts

Test continues on the other side of the sheet.

Unit 2 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

8. Independent tasks each succeed with probability 0.9. Which shortcut gives at least one failure among three tasks?	A. 1 minus the probability all three succeed. B. 0.9 plus 0.9 plus 0.9. C. 1 minus the probability exactly one succeeds. D. The probability all three fail. 10 pts
9. A sample is drawn without replacement from a box containing marked and unmarked cards. When must the sample contain a marked card?	A. When the sample size exceeds the number of unmarked cards. B. Whenever one marked card is present in the box. C. Only if every card in the box is marked. D. Whenever the sample size is less than the number of unmarked cards. 10 pts
10. X counts heads in three independent fair-coin tosses. Are 0, 1, 2, and 3 heads equally likely?	A. Yes; there are four possible counts. B. No; 0 is impossible. C. Yes; every fair-coin count is equally likely. D. No; 1 and 2 each occur through more sequences. 10 pts

27898

TEST PAGE 1

1.

C.

10 pts

2.

D.

10 pts

3.

C.

10 pts

4. A.

10 pts

5.

C.

10 pts

6. A.

10 pts

7.

D.

10 pts

TEST PAGE 2

8. A.

10 pts

9. A.

10 pts

10.

D.

10 pts

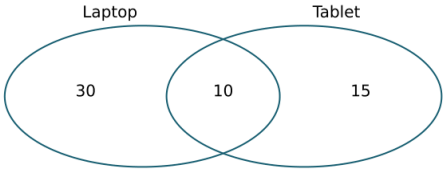
MSC 201 **46360**

Unit 2 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. Two independent retry procedures each have a constant chance of success. Which tends to have the shorter wait for the first success?	A. They always have the same typical wait. B. The one with the higher success chance per try. C. Whichever procedure happened to fail first. D. The one with the lower success chance per try. 10 pts
2. A model has one blank probability entry and lists every possible value. What does the blank need to account for?	A. Half the total probability, regardless of the other entries. B. All probability not already assigned to the other values. C. The probability of the most likely listed value. D. The average of the listed probabilities. 10 pts
3. A sample is drawn without replacement from a box containing marked and unmarked cards. When must the sample contain a marked card?	A. Only if every card in the box is marked. B. Whenever one marked card is present in the box. C. Whenever the sample size is less than the number of unmarked cards. D. When the sample size exceeds the number of unmarked cards. 10 pts
4. 40 students use laptops, 25 use tablets, and 10 use both. Why subtract 10 when counting either device? 	A. Adding the first two counts includes the overlap twice. B. The overlap does not use either device. C. Subtraction is required because the sample is random. D. Both-device users must be excluded entirely. 10 pts
5. A fair die is rolled. A means even; B means 6. Why are these events dependent?	A. Knowing B guarantees A, changing its probability. B. Both are separate die rolls. C. They cannot happen together. D. A fair die makes all events independent. 10 pts

Test continues on the other side of the sheet.

MSC 201 **46360**

Unit 2 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

6. Among 200 recorded jobs, 30 were late. What does 30/200 describe?	A. A guarantee that the next job is late. B. The observed late-job relative frequency. C. The exact long-run probability under every possible condition. D. The number of late jobs that must occur tomorrow. 10 pts
7. Probability is shifted from a low outcome to a higher outcome, with other probabilities unchanged. What happens to expected value?	A. It increases. B. It decreases. C. It must become the largest outcome. D. It cannot change. 10 pts
8. One student counts failed attempts before the first success. Another counts all attempts through the first success. How are their variables related?	A. The first includes one extra trial. B. Both counts are always equal. C. Neither count can be known when the first success occurs. D. The second includes one extra trial, the success itself. 10 pts
9. Half of tablet users use laptops, and half of non-tablet users use laptops. What does this suggest in these data?	A. Every student uses both devices. B. The events are mutually exclusive. C. Laptop use does not depend on tablet-use status. D. Every student uses exactly one device. 10 pts
10. Which shortcut finds at least one success in a fixed set of independent trials?	A. 1 minus the probability of no successes. B. The probability of exactly one success only. C. The expected success count. D. 1 minus the probability all succeed. 10 pts

46360

TEST PAGE 1

1.

B.

10 pts

2.

B.

10 pts

3.

D.

10 pts

4. A.

10 pts

5. A.

10 pts

TEST PAGE 2

6.

B.

10 pts

7. A.

10 pts

8.

D.

10 pts

9.

C.

10 pts

10. A.

10 pts

MSC 201 **85942**

Unit 2 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. To find P(tablet given laptop) from a table, which denominator is needed?	A. Only the number using neither. B. The number of laptop users. C. Only the number using both. D. The number of tablet users. 10 pts
2. Which design permits an exact binomial count model with independent randomized draws?	A. Replacing and remixing each drawn token. B. Drawing most of a small population without replacement. C. Always stopping at first red. D. Drawing without replacement until the box is empty. 10 pts
3. A supervisor predicts the expected total number of jobs across many sessions. What does that prediction describe?	A. The average total across repeated batches of sessions. B. The most jobs any batch could contain. C. The exact total in the next batch. D. The most likely total in every possible model. 10 pts
4. P(A)=0.30 and P(A given B)=0.30, with P(B)>0. What does this support?	A. B guarantees A. B. A and B are independent. C. A and B are mutually exclusive. D. A is impossible. 10 pts
5. A team redraws randomly from a mixed deck, replacing and remixing each card. It stops at the first heart. Which model describes the number of draws?	A. Binomial. B. Hypergeometric. C. Geometric. D. Continuous uniform. 10 pts
6. Which quantity naturally has discrete possible values?	A. The number of failed files. B. The exact mass of a package. C. An unrounded file-transfer duration. D. An unrounded measured length. 10 pts
7. A fair six-sided die is rolled. What is the probability of a result greater than 4?	A. 1/6. B. 2/6. C. 5/6. D. 4/6. 10 pts

Test continues on the other side of the sheet.

MSC 201 **85942**

Unit 2 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

8. Two successes in four trials can occur in several orders. Why must all those orders be considered?	A. The count event includes every order with two successes. B. Different orders are the same elementary outcome. C. The expected count is an order. D. Only the first successful order can occur. 10 pts
9. How do X>=3 and X>3 differ?	A. They are the same event. B. X>=3 requires four initial failures. C. X>=3 includes first success on trial 3. D. X>3 includes first success on trial 3. 10 pts
10. Given A, the chance of B is 0.7. What is the chance of not B given A?	A. 1.7. B. 0.7. C. Cannot be found unless A is independent of B. D. 0.3. 10 pts

85942

TEST PAGE 1

1.

B.

10 pts

2. A.

10 pts

3. A.

10 pts

4.

B.

10 pts

5.

C.

10 pts

6. A.

10 pts

7.

B.

10 pts

TEST PAGE 2

8. A.

10 pts

9.

C.

10 pts

10.

D.

10 pts

MSC 201 **46029**

Unit 2 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. Which Venn region is the complement of A OR B?	A. Only the overlap. B. Inside A only. C. Outside both circles. D. Every region inside either circle. 10 pts
2. In a fixed set of binomial trials, what bounds the number of successes?	A. There is no upper limit. B. The expected count is its upper limit. C. The number of failures is always its upper limit. D. The number of trials is its upper limit. 10 pts
3. Ten cables are sampled without replacement from 100,000. Why might binomial be a useful approximation?	A. The small sample barely changes the population composition. B. All samples have the same faulty count. C. The faulty count is continuous. D. No-replacement draws become exactly independent. 10 pts
4. For two trials with positive chance of two failures, how do exactly one failure and at least one failure compare?	A. The two events are complements. B. Exactly one has greater probability. C. They must be equal. D. At least one has greater probability. 10 pts
5. A game charges a fixed entry fee and sometimes awards a prize. What distinguishes net gain from the prize amount?	A. Net gain ignores outcomes with no prize. B. Net gain equals the largest possible prize. C. The fee changes the chance of each prize. D. Net gain accounts for the fee paid to play. 10 pts
6. On a die, A is even and B is greater than 3. Which results satisfy at least one condition?	A. 4, 6 only. B. All six results. C. 2, 4, 5, 6. D. 1, 3 only. 10 pts
7. In many sessions, 20% have no failures. How might this inform a model?	A. Use 0.20 as an empirical estimate for P(X=0). B. Declare zero failures impossible. C. Infer that every fifth session must have no failures. D. Set every count probability to 0.20. 10 pts

Test continues on the other side of the sheet.

MSC 201 **46029**

Unit 2 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

8. A teacher draws cards without replacement until the first marked card. Why is the draw number generally neither binomial nor geometric?	A. The draw number is continuous. B. A success has to be a desirable outcome. C. Stopping at the first success always makes the count binomial. D. The trial count is not fixed, and the chance can change between draws. 10 pts
9. X counts trials through first success; Y counts failures before it. What is the relationship?	A. X=Y. B. Y=X+1. C. X=Y+1. D. X=2Y. 10 pts
10. Tokens are replaced and mixed after each draw. Compared with leaving them out, what does replacement preserve?	A. A decreasing total number of tokens. B. The same token being selected every time. C. A guarantee that colors alternate. D. The original color probabilities on each independent draw. 10 pts

46029

TEST PAGE 1

1.

C.

10 pts

2.

D.

10 pts

3. A.

10 pts

4.

D.

10 pts

5.

D.

10 pts

6.

C.

10 pts

7. A.

10 pts

TEST PAGE 2

8.

D.

10 pts

9.

C.

10 pts

10.

D.

10 pts

MSC 201 **54217**

Unit 2 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. Which Venn region is the complement of A OR B?	A. Only the overlap. B. Outside both circles. C. Every region inside either circle. D. Inside A only. 10 pts
2. P(A)=0.30 and P(A given B)=0.30, with P(B)>0. What does this support?	A. A and B are independent. B. A and B are mutually exclusive. C. A is impossible. D. B guarantees A. 10 pts
3. In many sessions, 20% have no failures. How might this inform a model?	A. Use 0.20 as an empirical estimate for P(X=0). B. Declare zero failures impossible. C. Infer that every fifth session must have no failures. D. Set every count probability to 0.20. 10 pts
4. A buyer counts rejected items among a fixed number of independent inspections with the same rejection chance. What should be called a success for this model?	A. A set with no rejections. B. A rejection, because that is what the variable counts. C. An acceptance, because success must be favorable. D. A change in the rejection chance. 10 pts
5. A box contains marked and unmarked cards. An unmarked card is drawn and kept out. What happens to the chance that the next card is marked?	A. It stays the same. B. It increases. C. It decreases. D. It becomes zero. 10 pts
6. A service meets its deadline with probability 0.80. What is the probability of missing it?	A. 0.40. B. 0.20. C. 1.80. D. 0.80. 10 pts
7. To find the probability of A followed by B when the stages may be dependent, which expression fits?	A. P(A) plus P(B). B. P(A) times P(B given A). C. P(A) minus P(B given A). D. P(A) times P(B) without checking independence. 10 pts

Test continues on the other side of the sheet.

MSC 201 **54217**

Unit 2 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

8. Probability is shifted from a low outcome to a higher outcome, with other probabilities unchanged. What happens to expected value?	A. It decreases. B. It must become the largest outcome. C. It cannot change. D. It increases. 10 pts
9. Ten independent trials each have success probability 0.3. X counts successes. What makes a binomial model appropriate?	A. Every trial must succeed. B. Fixed trial count, two outcomes, independence, and constant success chance. C. Success probability exceeds 0.5. D. The expected count is an integer. 10 pts
10. How do X>=3 and X>3 differ?	A. They are the same event. B. X>3 includes first success on trial 3. C. X>=3 includes first success on trial 3. D. X>=3 requires four initial failures. 10 pts

54217

TEST PAGE 1

1.

B.

10 pts

2. A.

10 pts

3. A.

10 pts

4.

B.

10 pts

5.

B.

10 pts

6.

B.

10 pts

7.

B.

10 pts

TEST PAGE 2

8.

D.

10 pts

9.

B.

10 pts

10.

C.

10 pts

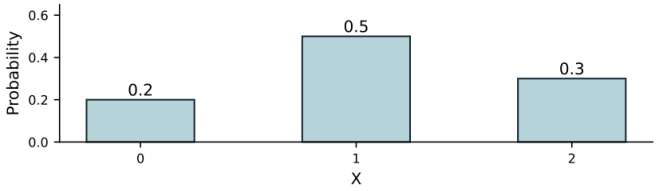
MSC 201 **22835**

Unit 2 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. X counts trials through first success. What does X>3 require?	A. At least one success in the first three trials. B. Exactly three successes. C. Failures on the first three trials. D. A success on trial 3. 10 pts
2. Among 200 recorded jobs, 30 were late. What does 30/200 describe?	A. The number of late jobs that must occur tomorrow. B. The observed late-job relative frequency. C. The exact long-run probability under every possible condition. D. A guarantee that the next job is late. 10 pts
3. A fixed-size sample is randomly selected from a batch of known good and faulty bulbs. Each selected bulb is set aside. Which model fits the number of good bulbs selected?	A. Binomial. B. Hypergeometric. C. Geometric. D. Continuous normal. 10 pts
4. Ten cables are sampled without replacement from 100,000. Why might binomial be a useful approximation?	A. No-replacement draws become exactly independent. B. All samples have the same faulty count. C. The faulty count is continuous. D. The small sample barely changes the population composition. 10 pts
5. A discrete probability table has a row at the cutoff. How do the events below the cutoff and at or below the cutoff differ? 	A. Only the first includes the row at the cutoff. B. They include opposite sides of the cutoff. C. Both exclude the row at the cutoff. D. Only the second includes the row at the cutoff. 10 pts
6. A game has negative expected net gain. Can a player still win money on one play?	A. No; every outcome must lose money. B. Yes, if positive-gain outcomes have positive probability. C. No; the largest prize must be negative. D. Only after the expected value changes. 10 pts

Test continues on the other side of the sheet.

MSC 201 **22835**

Unit 2 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. Among 20 tablet users, 12 also use laptops. What is P(laptop given tablet) in this sample?	A. 12 divided by all college students. B. 20/12. C. 8/20. D. 12/20. 10 pts
8. For two trials with positive chance of two failures, how do exactly one failure and at least one failure compare?	A. At least one has greater probability. B. Exactly one has greater probability. C. They must be equal. D. The two events are complements. 10 pts
9. Two successes in four trials can occur in several orders. Why must all those orders be considered?	A. Different orders are the same elementary outcome. B. Only the first successful order can occur. C. The expected count is an order. D. The count event includes every order with two successes. 10 pts
10. P(A)=0.30 and P(A given B)=0.30, with P(B)>0. What does this support?	A. B guarantees A. B. A and B are mutually exclusive. C. A and B are independent. D. A is impossible. 10 pts

22835

TEST PAGE 1

1.

C.

10 pts

2.

B.

10 pts

3.

B.

10 pts

4.

D.

10 pts

5.

D.

10 pts

6.

B.

10 pts

TEST PAGE 2

7.

D.

10 pts

8. A.

10 pts

9.

D.

10 pts

10.

C.

10 pts

MSC 201 **24139**

Unit 2 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. P(A OR B)=0.75. What is the probability of neither?	A. 0.75. B. 1.75. C. 0.50. D. 0.25. 10 pts
2. A game's expected prize is 4 dollars. Which entry fee makes expected net gain zero?	A. 8 dollars. B. 4 dollars. C. 0 dollars. D. The maximum possible prize regardless of its probability. 10 pts
3. A model assigns a 5% chance to 4 failures. One session has 4 failures. What follows?	A. All future sessions must have 4 failures. B. The model is disproved by this one allowed outcome. C. Its probability must now be 100%. D. The observation is uncommon but consistent with the model. 10 pts
4. X counts the successes. Which values satisfy fewer than 3 successes?	A. 0, 1, 2. B. 1, 2, 3 only. C. 3 and above. D. 0, 1, 2, 3. 10 pts
5. Plan A checks 10 items and counts successes. Plan B checks until the first success and counts trials. Which is geometric under independent constant-p trials?	A. Neither because trials are discrete. B. Both must be geometric. C. Plan B. D. Plan A. 10 pts
6. Six tokens are drawn without replacement from a small mixed box. What challenges an exact binomial model?	A. The count is an integer. B. There are only two colors. C. Draws change the composition and are dependent. D. The number of draws is fixed. 10 pts

Test continues on the other side of the sheet.

MSC 201 **24139**

Unit 2 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. An independent fair coin has just produced five tails. What is the chance of heads next?	A. Less than 1/2 because tails is favored. B. Greater than 1/2 because heads is due. C. 1. D. 1/2. 10 pts
8. The sample is an unordered set of cables. Why count combinations rather than different orders of the same set?	A. Selection order determines whether a cable is faulty. B. Every sample set has only one cable. C. Combinations assume replacement. D. Reordering the same selected cables does not create a new sample set. 10 pts
9. One student counts failed attempts before the first success. Another counts all attempts through the first success. How are their variables related?	A. The first includes one extra trial. B. Both counts are always equal. C. Neither count can be known when the first success occurs. D. The second includes one extra trial, the success itself. 10 pts
10. Half of tablet users use laptops, and half of non-tablet users use laptops. What does this suggest in these data?	A. Every student uses exactly one device. B. Every student uses both devices. C. Laptop use does not depend on tablet-use status. D. The events are mutually exclusive. 10 pts

24139

TEST PAGE 1

1.

D.

10 pts

2.

B.

10 pts

3.

D.

10 pts

4. A.

10 pts

5.

C.

10 pts

6.

C.

10 pts

TEST PAGE 2

7.

D.

10 pts

8.

D.

10 pts

9.

D.

10 pts

10.

C.

10 pts

MSC 201 **13829**

Unit 2 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. X counts trials through first success. What does X>3 require?	A. A success on trial 3. B. Exactly three successes. C. Failures on the first three trials. D. At least one success in the first three trials. 10 pts
2. A box contains marked and unmarked cards. An unmarked card is drawn and kept out. What happens to the chance that the next card is marked?	A. It stays the same. B. It increases. C. It decreases. D. It becomes zero. 10 pts
3. P(A)=0.4, P(B)=0.5, and their overlap is 0.2. What is P(A OR B)?	A. 1.1. B. 0.9. C. 0.7. D. 0.2. 10 pts
4. In many sessions, 20% have no failures. How might this inform a model?	A. Declare zero failures impossible. B. Use 0.20 as an empirical estimate for P(X=0). C. Set every count probability to 0.20. D. Infer that every fifth session must have no failures. 10 pts
5. A small prize is common and a large prize is rare. Why is the midpoint between the prizes generally a poor estimate of the expected payout?	A. Only the larger prize determines the mean. B. The expected payout must equal the larger prize. C. It gives the two prizes equal weight despite their different chances. D. The small prize does not count toward the average. 10 pts
6. To find P(tablet given laptop) from a table, which denominator is needed?	A. Only the number using both. B. The number of laptop users. C. Only the number using neither. D. The number of tablet users. 10 pts

Test continues on the other side of the sheet.

MSC 201 **13829**

Unit 2 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. One student counts failed attempts before the first success. Another counts all attempts through the first success. How are their variables related?	A. The second includes one extra trial, the success itself. B. Both counts are always equal. C. The first includes one extra trial. D. Neither count can be known when the first success occurs. 10 pts
8. Six tokens are drawn without replacement from a small mixed box. What challenges an exact binomial model?	A. The count is an integer. B. The number of draws is fixed. C. There are only two colors. D. Draws change the composition and are dependent. 10 pts
9. A proposed event probability is 1.15. What is wrong?	A. A probability cannot exceed 1. B. All probabilities must be integers. C. Only complements may exceed 1. D. Probabilities must always equal 0.5. 10 pts
10. Two mutually exclusive events each have positive probability. Can they be independent?	A. No; one occurring makes the other impossible. B. Yes; both have positive probabilities. C. Only if their probabilities add to 1. D. Yes; disjoint always means independent. 10 pts

13829

TEST PAGE 1

1.

C.

10 pts

2.

B.

10 pts

3.

C.

10 pts

4.

B.

10 pts

5.

C.

10 pts

6.

B.

10 pts

TEST PAGE 2

7. A.

10 pts

8.

D.

10 pts

9. A.

10 pts

10. A.

10 pts

MSC 201 **75511**

Unit 2 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. The success chance improves after each failed attempt because the process learns. Why is a geometric model questionable?	A. There is a stopping rule. B. The success probability is not constant. C. Failures are counted. D. The first success occurs eventually. 10 pts
2. A box contains marked and unmarked cards. An unmarked card is drawn and kept out. What happens to the chance that the next card is marked?	A. It becomes zero. B. It stays the same. C. It increases. D. It decreases. 10 pts
3. A proposed event probability is 1.15. What is wrong?	A. All probabilities must be integers. B. A probability cannot exceed 1. C. Only complements may exceed 1. D. Probabilities must always equal 0.5. 10 pts
4. A fair die is rolled. A means even; B means 6. Why are these events dependent?	A. Both are separate die rolls. B. Knowing B guarantees A, changing its probability. C. They cannot happen together. D. A fair die makes all events independent. 10 pts
5. To find P(tablet given laptop) from a table, which denominator is needed?	A. Only the number using both. B. The number of laptop users. C. The number of tablet users. D. Only the number using neither. 10 pts
6. A fixed-size sample is randomly selected from a batch of known good and faulty bulbs. Each selected bulb is set aside. Which model fits the number of good bulbs selected?	A. Geometric. B. Continuous normal. C. Hypergeometric. D. Binomial. 10 pts
7. Probability is shifted from a low outcome to a higher outcome, with other probabilities unchanged. What happens to expected value?	A. It cannot change. B. It must become the largest outcome. C. It increases. D. It decreases. 10 pts

Test continues on the other side of the sheet.

MSC 201 **75511**

Unit 2 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

8. Ten independent trials each have success probability 0.3. X counts successes. What makes a binomial model appropriate?	A. Success probability exceeds 0.5. B. Every trial must succeed. C. The expected count is an integer. D. Fixed trial count, two outcomes, independence, and constant success chance. 10 pts
9. For two trials with positive chance of two failures, how do exactly one failure and at least one failure compare?	A. The two events are complements. B. At least one has greater probability. C. Exactly one has greater probability. D. They must be equal. 10 pts
10. X counts heads in three independent fair-coin tosses. Are 0, 1, 2, and 3 heads equally likely?	A. Yes; every fair-coin count is equally likely. B. Yes; there are four possible counts. C. No; 0 is impossible. D. No; 1 and 2 each occur through more sequences. 10 pts

75511

TEST PAGE 1

1.

B.

10 pts

2.

C.

10 pts

3.

B.

10 pts

4.

B.

10 pts

5.

B.

10 pts

6.

C.

10 pts

7.

C.

10 pts

TEST PAGE 2

8.

D.

10 pts

9.

B.

10 pts

10.

D.

10 pts

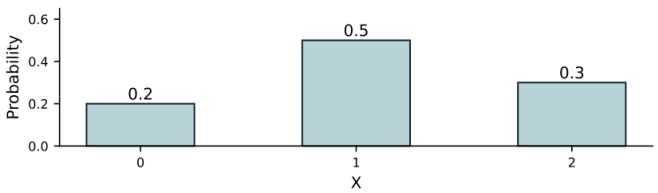
MSC 201 **14891**

Unit 2 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. A box has 3 red and 7 blue tokens; 5 are drawn without replacement. Why is 4 red impossible?	A. The sample contains 5 tokens. B. Half the population is drawn. C. Red probability must be zero. D. Only 3 red tokens exist. 10 pts
2. On a die, A is even and B is greater than 3. Which results satisfy both?	A. 2 only. B. 5 only. C. 2, 4, 5, 6. D. 4 and 6. 10 pts
3. A buyer counts rejected items among a fixed number of independent inspections with the same rejection chance. What should be called a success for this model?	A. A set with no rejections. B. An acceptance, because success must be favorable. C. A change in the rejection chance. D. A rejection, because that is what the variable counts. 10 pts
4. P(A)=0.30 and P(A given B)=0.30, with P(B)>0. What does this support?	A. A is impossible. B. A and B are mutually exclusive. C. B guarantees A. D. A and B are independent. 10 pts
5. Ten independent trials each have success probability 0.3. X counts successes. What makes a binomial model appropriate?	A. Fixed trial count, two outcomes, independence, and constant success chance. B. Every trial must succeed. C. Success probability exceeds 0.5. D. The expected count is an integer. 10 pts
6. A discrete probability table has a row at the cutoff. How do the events below the cutoff and at or below the cutoff differ? 	A. Both exclude the row at the cutoff. B. Only the second includes the row at the cutoff. C. They include opposite sides of the cutoff. D. Only the first includes the row at the cutoff. 10 pts

Test continues on the other side of the sheet.

MSC 201 **14891**

Unit 2 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. Among 20 tablet users, 12 also use laptops. What is P(laptop given tablet) in this sample?	A. 12/20. B. 20/12. C. 12 divided by all college students. D. 8/20. 10 pts
8. A geometric experiment has had several failures in a row. How does the next trial's chance of success compare with the initial chance?	A. It rises because a success is due. B. It stays the same. C. It drops because a failure streak has begun. D. It depends on the number of earlier failures. 10 pts
9. A game raises its fixed entry fee but leaves its prize distribution alone. What happens to the mean and SD of net gain?	A. Both rise. B. Both fall. C. The mean stays the same and the SD falls. D. The mean falls and the SD stays the same. 10 pts
10. A conditioning event has probability zero. What happens to the usual conditional-probability ratio?	A. Zero divided by zero equals zero. B. Its denominator is zero, so this ratio is undefined. C. The ratio automatically equals 0.5. D. The ratio automatically equals 1. 10 pts

14891

TEST PAGE 1

1.

D.

10 pts

2.

D.

10 pts

3.

D.

10 pts

4.

D.

10 pts

5. A.

10 pts

6.

B.

10 pts

TEST PAGE 2

7. A.

10 pts

8.

B.

10 pts

9.

D.

10 pts

10.

B.

10 pts

MSC 201 **96692**

Unit 2 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. A table lists every possible value, and each probability is nonnegative. What else must be true for the table to describe a probability distribution?	A. All values must have equal probabilities. B. The probabilities must rise as the values increase. C. The values must be consecutive whole numbers. D. The probabilities must sum to one. 10 pts
2. A sample is drawn without replacement from a box containing marked and unmarked cards. When must the sample contain a marked card?	A. Whenever the sample size is less than the number of unmarked cards. B. When the sample size exceeds the number of unmarked cards. C. Only if every card in the box is marked. D. Whenever one marked card is present in the box. 10 pts
3. A service meets its deadline with probability 0.80. What is the probability of missing it?	A. 1.80. B. 0.80. C. 0.20. D. 0.40. 10 pts
4. The mean waiting count is 4 trials. Must the first success occur by trial 4?	A. No; longer waits remain possible. B. Yes; the mean is a deadline. C. Yes; 4 is the maximum. D. No; it must occur exactly on trial 5. 10 pts
5. A binomial count has mean 6. What does this imply about the next experiment?	A. Its count may differ from 6. B. Its count cannot exceed 6. C. Its count must be exactly 6. D. Its count cannot be below 6. 10 pts
6. P(A OR B)=0.75. What is the probability of neither?	A. 1.75. B. 0.75. C. 0.25. D. 0.50. 10 pts
7. Two distinct terminal paths describe exactly one head in two tosses. How is their combined probability found?	A. Multiply the two path probabilities. B. Add the two path probabilities. C. Ignore one because both describe one head. D. Subtract the smaller from the larger. 10 pts

Test continues on the other side of the sheet.

MSC 201 **96692**

Unit 2 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

8. A small prize is common and a large prize is rare. Why is the midpoint between the prizes generally a poor estimate of the expected payout?	A. The small prize does not count toward the average. B. Only the larger prize determines the mean. C. It gives the two prizes equal weight despite their different chances. D. The expected payout must equal the larger prize. 10 pts
9. One student counts failed attempts before the first success. Another counts all attempts through the first success. How are their variables related?	A. The second includes one extra trial, the success itself. B. Neither count can be known when the first success occurs. C. The first includes one extra trial. D. Both counts are always equal. 10 pts
10. P(A)=0.4, P(B)=0.5, and P(A AND B)=0.2. Which check supports independence?	A. 0.5 minus 0.4 equals 0.1. B. 0.4 times 0.5 equals 0.2. C. 0.4 plus 0.5 equals 0.9. D. The intersection is positive. 10 pts

96692

TEST PAGE 1

1.

D.

10 pts

2.

B.

10 pts

3.

C.

10 pts

4. A.

10 pts

5. A.

10 pts

6.

C.

10 pts

7.

B.

10 pts

TEST PAGE 2

8.

C.

10 pts

9. A.

10 pts

10.

B.

10 pts

MSC 201 **83239**

Unit 2 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. Among 200 recorded jobs, 30 were late. What does 30/200 describe?	A. A guarantee that the next job is late. B. The number of late jobs that must occur tomorrow. C. The exact long-run probability under every possible condition. D. The observed late-job relative frequency. 10 pts
2. Tokens are replaced and mixed after each draw. Compared with leaving them out, what does replacement preserve?	A. The original color probabilities on each independent draw. B. The same token being selected every time. C. A decreasing total number of tokens. D. A guarantee that colors alternate. 10 pts
3. A conditioning event has probability zero. What happens to the usual conditional-probability ratio?	A. Its denominator is zero, so this ratio is undefined. B. The ratio automatically equals 1. C. Zero divided by zero equals zero. D. The ratio automatically equals 0.5. 10 pts
4. A payout variable is measured in dollars. What units belong to its variance?	A. Percent. B. Dollars. C. No units. D. Dollars squared. 10 pts
5. A probability table totals 1 but contains one entry of -0.1. What follows?	A. It is valid because only the total matters. B. It is invalid because probabilities cannot be negative. C. It is valid if the mean is positive. D. Negative probabilities simply indicate rare events. 10 pts
6. Four cables are sampled uniformly without replacement from 12, of which 5 are faulty. Which model directly fits the faulty count?	A. Normal individual measurement. B. Hypergeometric. C. Geometric waiting count. D. Continuous uniform. 10 pts

Test continues on the other side of the sheet.

MSC 201 **83239**

Unit 2 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. A supplied probability table lists P(X=0),...,P(X=8). To find P(X<=2), which entries are needed?	A. The entries for 0, 1, and 2. B. Only the entry for 2. C. The entries for 3 through 8. D. The expected value. 10 pts
8. Among 20 tablet users, 12 also use laptops. What is P(laptop given tablet) in this sample?	A. 12 divided by all college students. B. 8/20. C. 20/12. D. 12/20. 10 pts
9. Independent trials have a constant chance of success strictly between zero and one. Which variable can take arbitrarily large values?	A. The number of outcomes on one yes-or-no trial. B. The number of successes in a fixed set of trials. C. The number of marked items in a fixed-size sample. D. The number of trials through the first success. 10 pts
10. Two independent retry procedures each have a constant chance of success. Which tends to have the shorter wait for the first success?	A. The one with the higher success chance per try. B. They always have the same typical wait. C. Whichever procedure happened to fail first. D. The one with the lower success chance per try. 10 pts

83239

TEST PAGE 1

1.

D.

10 pts

2. A.

10 pts

3. A.

10 pts

4.

D.

10 pts

5.

B.

10 pts

6.

B.

10 pts

TEST PAGE 2

7. A.

10 pts

8.

D.

10 pts

9.

D.

10 pts

10. A.

10 pts

MSC 201 **51596**

Unit 2 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. Device-use events appear independent in one class's table. What conclusion needs caution?	A. Their rates match within the observed table. B. The table records that class. C. They are independent for all students at the college. D. The comparison used conditional rates. 10 pts
2. X counts trials through first success. What does X>3 require?	A. Exactly three successes. B. At least one success in the first three trials. C. Failures on the first three trials. D. A success on trial 3. 10 pts
3. On a die, A is even and B is greater than 3. Which results satisfy at least one condition?	A. 1, 3 only. B. All six results. C. 2, 4, 5, 6. D. 4, 6 only. 10 pts
4. Which event is the complement of at least one failure in three trials?	A. No failures. B. At least two failures. C. Exactly one failure. D. All three failures. 10 pts
5. A team redraws randomly from a mixed deck, replacing and remixing each card. It stops at the first heart. Which model describes the number of draws?	A. Hypergeometric. B. Continuous uniform. C. Binomial. D. Geometric. 10 pts
6. Two mutually exclusive events each have positive probability. Can they be independent?	A. Only if their probabilities add to 1. B. Yes; both have positive probabilities. C. Yes; disjoint always means independent. D. No; one occurring makes the other impossible. 10 pts
7. X counts broken cables in independent checks. Can broken be called success in a binomial model?	A. Only if more than half are broken. B. Yes; success means the counted outcome. C. No; success must be desirable. D. Only if no cables are broken. 10 pts

Test continues on the other side of the sheet.

MSC 201 **51596**

Unit 2 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

8. A game raises its fixed entry fee but leaves its prize distribution alone. What happens to the mean and SD of net gain?	A. The mean falls and the SD stays the same. B. Both fall. C. Both rise. D. The mean stays the same and the SD falls. 10 pts
9. In many sessions, 20% have no failures. How might this inform a model?	A. Set every count probability to 0.20. B. Infer that every fifth session must have no failures. C. Declare zero failures impossible. D. Use 0.20 as an empirical estimate for P(X=0). 10 pts
10. Ten cables are sampled without replacement from 100,000. Why might binomial be a useful approximation?	A. The small sample barely changes the population composition. B. All samples have the same faulty count. C. The faulty count is continuous. D. No-replacement draws become exactly independent. 10 pts

51596

TEST PAGE 1

1.

C.

10 pts

2.

C.

10 pts

3.

C.

10 pts

4. A.

10 pts

5.

D.

10 pts

6.

D.

10 pts

7.

B.

10 pts

TEST PAGE 2

8. A.

10 pts

9.

D.

10 pts

10. A.

10 pts

MSC 201 **28843**

Unit 2 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. X counts broken cables in independent checks. Can broken be called success in a binomial model?	A. Only if no cables are broken. B. No; success must be desirable. C. Yes; success means the counted outcome. D. Only if more than half are broken. 10 pts
2. P(A)=0.4, P(B)=0.5, and their overlap is 0.2. What is P(A OR B)?	A. 1.1. B. 0.9. C. 0.2. D. 0.7. 10 pts
3. A small prize is common and a large prize is rare. Why is the midpoint between the prizes generally a poor estimate of the expected payout?	A. It gives the two prizes equal weight despite their different chances. B. The small prize does not count toward the average. C. The expected payout must equal the larger prize. D. Only the larger prize determines the mean. 10 pts
4. Which Venn region is the complement of A OR B?	A. Only the overlap. B. Outside both circles. C. Every region inside either circle. D. Inside A only. 10 pts
5. Which sequence gives the first success on trial 4?	A. Four successes. B. Failure, success, failure, success. C. Success, failure, failure, failure. D. Failure, failure, failure, success. 10 pts
6. An inspector checks a fixed batch of independent items with the same defect chance. Which variable fits a binomial model?	A. The attempt on which the first defect is found. B. The exact mass of the first item. C. The average time needed to check an item. D. The number of defective items found. 10 pts
7. A service meets its deadline with probability 0.80. What is the probability of missing it?	A. 1.80. B. 0.40. C. 0.20. D. 0.80. 10 pts

Test continues on the other side of the sheet.

MSC 201 **28843**

Unit 2 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

8. Two mixed populations have the same size, but one has a larger share of faulty cables. Both are sampled randomly without replacement at the same sample size. How do their expected faulty counts compare?	A. The expected count is higher for the population with a larger faulty share. B. The expected counts are equal because the sample sizes agree. C. The expected counts depend only on whether order is recorded. D. The expected count is lower for the population with a larger faulty share. 10 pts
9. A fair die is rolled. A means even; B means 6. Why are these events dependent?	A. They cannot happen together. B. A fair die makes all events independent. C. Both are separate die rolls. D. Knowing B guarantees A, changing its probability. 10 pts
10. A model assigns a 5% chance to 4 failures. One session has 4 failures. What follows?	A. Its probability must now be 100%. B. The model is disproved by this one allowed outcome. C. The observation is uncommon but consistent with the model. D. All future sessions must have 4 failures. 10 pts

28843

TEST PAGE 1

1.

C.

10 pts

2.

D.

10 pts

3. A.

10 pts

4.

B.

10 pts

5.

D.

10 pts

6.

D.

10 pts

7.

C.

10 pts

TEST PAGE 2

8. A.

10 pts

9.

D.

10 pts

10.

C.

10 pts

MSC 201 **14838**

Unit 2 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. Two mixed populations have the same size, but one has a larger share of faulty cables. Both are sampled randomly without replacement at the same sample size. How do their expected faulty counts compare?	A. The expected count is lower for the population with a larger faulty share. B. The expected counts are equal because the sample sizes agree. C. The expected counts depend only on whether order is recorded. D. The expected count is higher for the population with a larger faulty share. 10 pts
2. A table contains 5 students using both devices. What does this show about mutual exclusivity?	A. Neither event has positive probability. B. The two use events are not mutually exclusive. C. They must be independent. D. They are mutually exclusive. 10 pts
3. A teacher draws cards without replacement until the first marked card. Why is the draw number generally neither binomial nor geometric?	A. A success has to be a desirable outcome. B. The trial count is not fixed, and the chance can change between draws. C. Stopping at the first success always makes the count binomial. D. The draw number is continuous. 10 pts
4. A geometric experiment has had several failures in a row. How does the next trial's chance of success compare with the initial chance?	A. It stays the same. B. It rises because a success is due. C. It drops because a failure streak has begun. D. It depends on the number of earlier failures. 10 pts
5. One game always pays its expected payout. Another has the same expected payout but offers several possible payouts. How do they differ? 	A. Both games must have zero spread. B. Only the first game has positive spread. C. Their spreads must match because their means match. D. Only the second game has positive spread. 10 pts

Test continues on the other side of the sheet.

MSC 201 **14838**

Unit 2 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

6. A conditioning event has probability zero. What happens to the usual conditional-probability ratio?	A. Zero divided by zero equals zero. B. The ratio automatically equals 0.5. C. Its denominator is zero, so this ratio is undefined. D. The ratio automatically equals 1. 10 pts
7. P(A)=0.30 and P(A given B)=0.30, with P(B)>0. What does this support?	A. A and B are independent. B. A and B are mutually exclusive. C. A is impossible. D. B guarantees A. 10 pts
8. Which shortcut finds at least one success in a fixed set of independent trials?	A. The probability of exactly one success only. B. 1 minus the probability of no successes. C. 1 minus the probability all succeed. D. The expected success count. 10 pts
9. Which quantity naturally has discrete possible values?	A. An unrounded measured length. B. An unrounded file-transfer duration. C. The number of failed files. D. The exact mass of a package. 10 pts
10. On a die, A is even and B is greater than 3. Which results satisfy at least one condition?	A. 1, 3 only. B. All six results. C. 2, 4, 5, 6. D. 4, 6 only. 10 pts

14838

TEST PAGE 1

1.

D.

10 pts

2.

B.

10 pts

3.

B.

10 pts

4. A.

10 pts

5.

D.

10 pts

TEST PAGE 2

6.

C.

10 pts

7. A.

10 pts

8.

B.

10 pts

9.

C.

10 pts

10.

C.

10 pts

MSC 201 **80696**

Unit 2 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. Among 20 tablet users, 12 also use laptops. What is P(laptop given tablet) in this sample?	A. 12 divided by all college students. B. 12/20. C. 20/12. D. 8/20. 10 pts
2. A machine's success chance steadily falls as it warms. Which binomial requirement is questionable?	A. There being two possible outcomes. B. Constant success probability across trials. C. A fixed number of trials. D. A numerical success count. 10 pts
3. Which design permits an exact binomial count model with independent randomized draws?	A. Drawing without replacement until the box is empty. B. Always stopping at first red. C. Drawing most of a small population without replacement. D. Replacing and remixing each drawn token. 10 pts
4. One student counts failed attempts before the first success. Another counts all attempts through the first success. How are their variables related?	A. Both counts are always equal. B. Neither count can be known when the first success occurs. C. The first includes one extra trial. D. The second includes one extra trial, the success itself. 10 pts
5. A table lists every possible value, and each probability is nonnegative. What else must be true for the table to describe a probability distribution?	A. All values must have equal probabilities. B. The probabilities must rise as the values increase. C. The probabilities must sum to one. D. The values must be consecutive whole numbers. 10 pts
6. 20 students use both devices; 40 use tablets and 80 use laptops. Which conditional probability is larger?	A. Neither can use the overlap count. B. Tablet use given laptop use. C. They must be equal. D. Laptop use given tablet use. 10 pts

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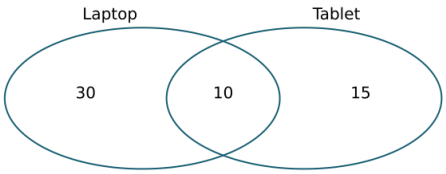
MSC 201 **80696**

Unit 2 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. 40 students use laptops, 25 use tablets, and 10 use both. Why subtract 10 when counting either device?  	A. Subtraction is required because the sample is random. B. The overlap does not use either device. C. Adding the first two counts includes the overlap twice. D. Both-device users must be excluded entirely. 10 pts
8. A proposed event probability is 1.15. What is wrong?	A. A probability cannot exceed 1. B. Probabilities must always equal 0.5. C. Only complements may exceed 1. D. All probabilities must be integers. 10 pts
9. A geometric experiment has had several failures in a row. How does the next trial's chance of success compare with the initial chance?	A. It depends on the number of earlier failures. B. It stays the same. C. It rises because a success is due. D. It drops because a failure streak has begun. 10 pts
10. A count variable has expected value 0.4. What does that mean?	A. Its long-run average can be 0.4 despite integer observations. B. Exactly 40% of sessions must contain one event. C. A single session must contain 0.4 events. D. The model is invalid. 10 pts

80696

TEST PAGE 1

1.

B.

10 pts

2.

B.

10 pts

3.

D.

10 pts

4.

D.

10 pts

5.

C.

10 pts

6.

D.

10 pts

TEST PAGE 2

7.

C.

10 pts

8. A.

10 pts

9.

B.

10 pts

10. A.

10 pts

MSC 201 **42390**

Unit 2 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. A proposed event probability is 1.15. What is wrong?	A. Probabilities must always equal 0.5. B. Only complements may exceed 1. C. A probability cannot exceed 1. D. All probabilities must be integers. 10 pts
2. A device is tested a fixed number of times, but each repair after a failed trial improves its chance of passing. Why is a simple binomial model questionable?	A. The result is a count. B. The number of trials is fixed. C. The test records successes and failures. D. The pass chance depends on previous outcomes. 10 pts
3. X counts heads in three independent fair-coin tosses. Are 0, 1, 2, and 3 heads equally likely?	A. Yes; there are four possible counts. B. Yes; every fair-coin count is equally likely. C. No; 0 is impossible. D. No; 1 and 2 each occur through more sequences. 10 pts
4. Plan A checks 10 items and counts successes. Plan B checks until the first success and counts trials. Which is geometric under independent constant-p trials?	A. Plan B. B. Neither because trials are discrete. C. Plan A. D. Both must be geometric. 10 pts
5. Ten independent trials each have success probability 0.3. X counts successes. What makes a binomial model appropriate?	A. Every trial must succeed. B. Fixed trial count, two outcomes, independence, and constant success chance. C. The expected count is an integer. D. Success probability exceeds 0.5. 10 pts
6. Which event is the complement of at least one failure in three trials?	A. At least two failures. B. All three failures. C. No failures. D. Exactly one failure. 10 pts
7. An independent fair coin has just produced five tails. What is the chance of heads next?	A. Less than 1/2 because tails is favored. B. 1. C. 1/2. D. Greater than 1/2 because heads is due. 10 pts

Test continues on the other side of the sheet.

MSC 201 **42390**

Unit 2 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

8. A payout variable is measured in dollars. What units belong to its variance?	A. Dollars squared. B. Percent. C. No units. D. Dollars. 10 pts
9. Two mixed populations have the same size, but one has a larger share of faulty cables. Both are sampled randomly without replacement at the same sample size. How do their expected faulty counts compare?	A. The expected count is higher for the population with a larger faulty share. B. The expected counts depend only on whether order is recorded. C. The expected count is lower for the population with a larger faulty share. D. The expected counts are equal because the sample sizes agree. 10 pts
10. Two distinct terminal paths describe exactly one head in two tosses. How is their combined probability found?	A. Subtract the smaller from the larger. B. Multiply the two path probabilities. C. Ignore one because both describe one head. D. Add the two path probabilities. 10 pts

42390

TEST PAGE 1

1.

C.

10 pts

2.

D.

10 pts

3.

D.

10 pts

4. A.

10 pts

5.

B.

10 pts

6.

C.

10 pts

7.

C.

10 pts

TEST PAGE 2

8. A.

10 pts

9. A.

10 pts

10.

D.

10 pts

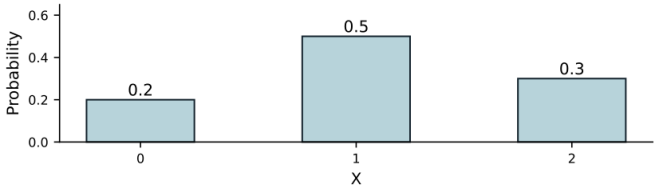
MSC 201 **12607**

Unit 2 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. Which Venn region is the complement of A OR B?	A. Inside A only. B. Outside both circles. C. Only the overlap. D. Every region inside either circle. 10 pts
2. A team redraws randomly from a mixed deck, replacing and remixing each card. It stops at the first heart. Which model describes the number of draws?	A. Continuous uniform. B. Binomial. C. Geometric. D. Hypergeometric. 10 pts
3. In a fixed set of binomial trials, what bounds the number of successes?	A. The expected count is its upper limit. B. The number of failures is always its upper limit. C. The number of trials is its upper limit. D. There is no upper limit. 10 pts
4. X counts trials through first success. What does X>3 require?	A. At least one success in the first three trials. B. Failures on the first three trials. C. Exactly three successes. D. A success on trial 3. 10 pts
5. A discrete probability table has a row at the cutoff. How do the events below the cutoff and at or below the cutoff differ? 	A. Both exclude the row at the cutoff. B. Only the second includes the row at the cutoff. C. Only the first includes the row at the cutoff. D. They include opposite sides of the cutoff. 10 pts
6. A box contains both red and blue tokens. After a red token is drawn without replacement, how does the next draw's chance of red change?	A. It always falls to zero. B. It stays the same. C. It increases. D. It decreases. 10 pts

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MSC 201 **12607**

Unit 2 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. P(A OR B)=0.75. What is the probability of neither?	A. 0.75. B. 0.50. C. 0.25. D. 1.75. 10 pts
8. A fair coin gives 8 heads in 10 tosses. What follows?	A. The next two tosses must be tails. B. The coin's future head probability is now 0.8. C. Such a short-run imbalance can occur with a fair coin. D. Fairness requires exactly 5 heads in every ten tosses. 10 pts
9. Four cables are sampled uniformly without replacement from 12, of which 5 are faulty. Which model directly fits the faulty count?	A. Hypergeometric. B. Geometric waiting count. C. Continuous uniform. D. Normal individual measurement. 10 pts
10. A payout variable is measured in dollars. What units belong to its variance?	A. Dollars. B. Dollars squared. C. No units. D. Percent. 10 pts

12607

TEST PAGE 1

1.

B.

10 pts

2.

C.

10 pts

3.

C.

10 pts

4.

B.

10 pts

5.

B.

10 pts

6.

D.

10 pts

TEST PAGE 2

7.

C.

10 pts

8.

C.

10 pts

9. A.

10 pts

10.

B.

10 pts